

the bleed 0.00



clear

start the room ▶

one slider sours the whole page – colour, grain, aberration, the hum, and her voice, together. Returns to spotless ash at 0.

the wrong hour – the dictionary & her voice

a live parts bin for wuld.ink · Web Audio + Canvas · vanilla, single-file, offline · v2 – revised against the shipped wrong-hour.js (K211) + her original game voices, ported byte-for-byte from Sfx.gd

Read this first. Everything below is **audible or visible and copy-pasteable** – click, hear it, hit source, take the code. Plain JS, no libraries, no CDN, no network, no localStorage; it runs off a file://. This is v2: the sounds now **mirror the values that actually ship** in src/components/wrong-hour.js, so what you copy is what the site already plays – each demo names its shipped symbol like this: → wrong-hour.js: bell(). Two laws carry over from the game, unchanged:

1. Synthesize, don't sample. No .wav, ever – same rule as Sfx.gd. Oscillators, noise buffers, a WaveShaper for grit. Smaller, tunable, reactive.

2. One dial, colour-only. A single 0..1 – on the site $\text{rot()} = \text{vfx} \times (0.4 + 0.6 \cdot \text{bleedNow}())$ – sours everything and never darkens to black: lightness is held, only saturation and hue drift toward red. At 0 the site is spotless ash: the false relief the night later rots. In sound the mirror-law is **sour timbre / space / density, never mute to silence**.

The five-minute gotcha: a browser makes no sound until the user clicks. AudioContext starts suspended; you `ctx.resume()` inside a real gesture. Every button here does that; the site does it once via `whenReady/arm`.

A · Sound – the cue library & the bed

Bias for "the wrong hour": restraint. Dim, a little too slow, the hum a touch too loud. Inharmonic = unease. Sub-bass = dread you feel, not hear. Envelope matters more than waveform. Nothing resolves to a bright major chord – it is never triumphant. The cues below are the exact eight in CUES{}; the bed is the four-mood ambience engine.

A0 · The spine – one context, one room

Make the AudioContext once, unlock on first gesture, route everything through a shared room (a gentle lowpass + a synthesized-IR reverb) so every sound sits in the same dim space. Build this first; every cue connects to room.

one tap, in the empty room

the click is mostly its own reverb tail – that's the room doing the work.

[spine source](#) ▾ [the synth reverb](#) ▾

A1 · boot `power-on` → wrong-hour.js: bootSeq()

A computer waking – a rounded sub-whoomp + a warm mains-hum that opens under a rising lowpass + soft air + a breathing CRT shimmer. Deliberately ASMR, no clanky beep. Fires once on first load, and on data-wh="boot" / power scenes.

power on flash + the boot sequence together.

[source](#) ▾

A2 · thunk `degauss` → wrong-hour.js: thunk()

The iconic CRT whoomp: a sine dropping 150→38 Hz in 0.22 s + a short bandpassed coil buzz at 120 Hz. Page load, a heavy threshold, the site "powering on."

degauss

[source](#) ▾

A3 · key `the reader / typing` → wrong-hour.js: key()

A mechanical thock – a rounded ~120–175 Hz body + a soft bandpassed top-click + per-press variation, and a 12%-of-the-time faint spring. Fire one per keystroke; the variation keeps a fast typist from sounding like a machine gun.

key x1

a sentence of keys

[source](#) ▾

A4 · **soft** `navigation` → `wrong-hour.js: soft()`

A warm, muted analog pluck: a sawtooth through a resonant lowpass that snaps closed toward the fundamental (never bright), + a sub-octave for body. It walks an A-minor-pentatonic set (NAV_HZ) so consecutive nav clicks make a little falling melody. (K225 flags this one as maybe-silenced – Josiah's call.)

nav x1

walk the scale

[source](#) ▼

A5 · **click** `buttons` → `wrong-hour.js: click()`

A soft tick with a small round 260 Hz body – the plain button cue, distinct from the nav pluck and the key thock.

click

[source](#) ▼

A6 · **bell** `a chime grief got into` → `wrong-hour.js: bell()`

The centrepiece cue. Partial at inharmonic ratios [1, 2.76, 5.40, 8.93] over base 110 Hz, each with its own decay, + a sub-octave for weight, a big convolver tail, and a feedback echo. And it **bitcrushes as the room rots**: past `rot() > 0.25` it routes through a WaveShaper whose bit-depth falls as the bleed rises. Move the console slider, then strike it – the same chime, curdling. Use it once, at a threshold.

strike strike at bleed 0, then at 0.8 – hear the crush.

[source](#) ▼ [the crush curve](#) ▼

A7 · **purr** `the cat that lets itself in` → `wrong-hour.js: purr()`

A 55 Hz sine throbbled by a 22 Hz amplitude LFO. Warmth you feel more than hear. **Honest limit**: laptop speakers won't reproduce 55 Hz – headphones / felt register only. Never the sole cue.

purr

[source](#) ▼

A8 · word `"word landed"` → `wrong-hour.js: wordChime()`

A soft bright triangle chime walking a high pentatonic (WORD_HZ, G4-E5) + a shimmer partial. Fires on each completed word in the notes editor. The one place the palette is allowed a little sweetness – because it's earned, one word at a time.

word a few words

[source](#) ▾

A9 · the bed `the ambience engine` → MOODS · `ambBuildDrone` · AMB_TEX

The generative BED under the site: a detuned drone through a resonant lowpass wobbled by a slow "whir" LFO, + faint air, + a slow breath on the master gain, + sparse non-repeating piano motifs, + an occasional texture. It reads a per-mood config (room / clinical / oceanic / breeze – breeze is the shipped default). Pick a mood; start the drone; fire its texture; toggle the piano. This is the "adapt per state" hook: `map game/site-state → a mood, or interpolate the fields.`

mood: room clinical oceanic breeze ✓

start the bed ▶

fire this mood's texture

piano: off

breeze – the deep Radigue-ish drone + the aching "howl".

[the mood configs](#) ▾ [the drone](#) ▾ [the howl \(breeze\)](#) ▾ [the piano scheduler](#) ▾

A10 · reverse-swell `dread onset` **NEW**

Not in the shipped engine – offered for the terminal/threshold moments. A band of noise swelling backwards into a strike (the room inhaling before the bell). Pre-echo is the oldest trick for "something is about to happen"; keep it rare.

swell → strike

[source](#) ▾

A11 · tape wow/flutter `the dated machine` **NEW**

Also new: a slow pitch drift (wow) + a faster shallow warble (flutter) over a held tone – the sound of a machine that has been running too long, at the

wrong hour. Wrap it around a bed or a held note; scale the depth with the bleed for a tape that degrades as it rots.

held tone, worn

[source](#) ▾

Sound principles (the short version). Inharmonic partials → unease without a jumpscare. Sub-bass → dread. Envelope > waveform. Detune and drift so nothing is mechanically perfect (36 vs 36.4 Hz; 3.01 partials). Sparseness and silence do more than density. Never resolve bright – the day is not a reward, it's a shift change. And the bleed sours timbre, never mutes to silence.

B · Visual – the CRT read & the bleed

Ordinary at rest; the effect is the exception, not the constant. One rot() drives all of it, and it only ever moves colour – the tint layer is mix-blend-mode: color so lightness never budges.

B0 · The dial the centerpiece → rot() · bleedNow() · fxi()

Three numbers run everything. bleedNow() is the clock (0 at noon → 1 near 3am). rot() = $vfx \times (0.4 + 0.6 \cdot \text{bleedNow}())$ is the per-frame intensity for the always-on layers (tint, grain, aberration, vignette). fxi() = $0.4 + 0.6 \cdot \text{bleedNow}()$ is the swell for interaction one-shots. The console slider here stands in for rot(). On a game, replace the clock with a tension scalar and the whole system escalates on cue.

drive it from the top console – text, links, borders, buttons, grain and her voice sour together.

[the palette function](#) ▾

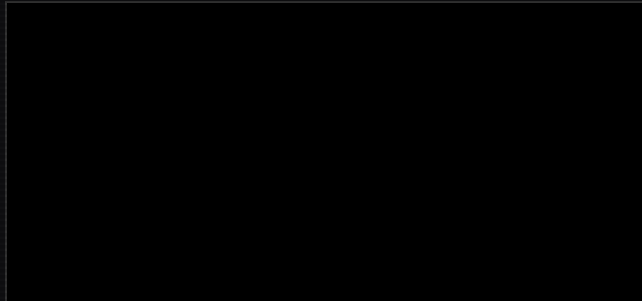
B1 · scanlines + vignette free CRT

Two CSS overlays – a repeating-linear-gradient for scanlines, a radial-gradient for the vignette – both position:fixed; pointer-events:none. This page wears them now. Zero JS, near-zero cost.

[the CSS](#) ▾

B2 · grain / ash `the air in the room` → `makeNoise()`

The shipped grain is real film grain: sparse light/dark speckles drawn once onto a transparent canvas and re-seeded, not a full per-pixel wash (cheaper, and it reads as dust in the beam rather than TV snow). Density + opacity ride `rot()`; below `rot ≤ 0.02` it skips the work entirely.

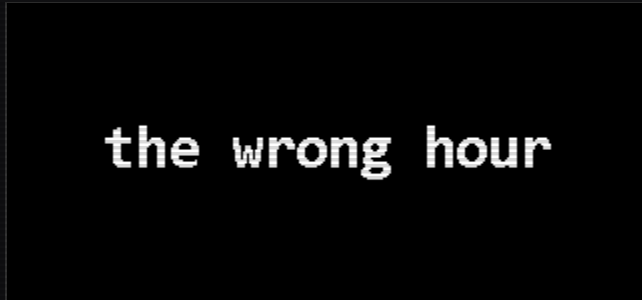


sparse speckle, re-seeded.
density rides the bleed.

[source](#) ▼

B3 · chromatic aberration `tired glass`

Split the red and blue channels a pixel or two apart, added with `globalCompositeOperation='lighter'`. Offset scales with **the bleed** – spotless at rest, fraying as it rots.



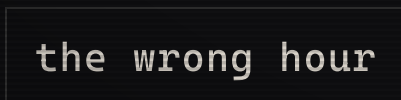
offset = bleed × 4px.
drive the console slider.

[source](#) ▼

B4 · the VFX registry `interaction one-shots` → `VFX · whFx() · fxFire()`

Interaction one-shots on a single stage, each peak swelled by `fxi()`: nav → pulse (accent tint), buttons → flick (brightness), typing → glitch (heading shiver), + burst / roll / bloom. Legibility-safe (blend holds lightness), off under prefers-reduced-motion, routed per element with `data-wh-vfx`.

`pulse` `flick` `glitch` `burst` `roll` `bloom`



[the one-shot CSS](#) ▼

B5 · the strangeness engine

session-age escalation

→ corruption() · intrudeWord ·

corruptTextNode

A ramp over tab-open time (corruption(), ~30 min). As it rises the page drifts: dread-words ghost near the pointer, a single character flickers to another glyph and heals, phantom link-targets appear – all gated by vfx>0 && !reduced-motion, throttled, and it never mutates the real text or nav. Drive it off game-time/tension instead of tab-age and the site "wears" on the reader.

ghost a word

flicker a glyph

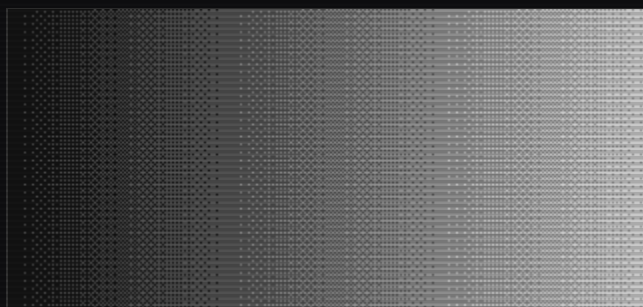
the reading room is open. it usually is, at this hour.

[source](#) ▾

B6 · dither / posterize

PS1 colour depth

Crush a smooth gradient to a few levels through an ordered 4×4 Bayer matrix – the limited-palette look, no banding. The same trick posterizes any canvas to the red/ash ramp.



3 levels, Bayer 4×4.

[source](#) ▾

B7 · phosphor text

the reader

Typewriter reveal + an occasional single-character corruption that heals + a soft glow. Glitch rate rides the bleed; each non-space fires a key().

reveal a line

[source](#) ▾

B8 · halation

bloom around light text

NEW

A cheap tell for "too bright for the tube": bright text bleeds a soft glow whose spread and red-shift grow with the bleed – a CRT that can't quite contain its own highlights. Pure text-shadow, driven by the dial.

toggle halation on this line

CIRCULATION: ONE. READERSHIP: ASSUMED.

[source](#) ▾

Visual principles. Colour-only degradation – the bleed sours, never dims to black. Restraint – at rest it's an ordinary dim page. One param, so the site sours as a single organism. Cheap-first – CSS scanlines/vignette/grain before a shader; spend WebGL only on true per-pixel warp. And with JS off there are **zero** effects and the page is byte-identical (the homepage invariant).

D · Her voice – the animalese dialect from the night

The desk-Yūrei assistant ships **zero audio by design** – but her voice already exists: the game synthesizes it in Sfx.gd, no samples. **D1 below is that original, ported byte-for-byte** (build the same 22050 Hz buffer sample-by-sample, so it's her actual timbre, not a look-alike). Under it: how to put her on the one dial, and a text-articulated expansion for the desk voice-lab – clearly marked as not-in-the-game.

D1 · her voice, as the game makes it the canon → Sfx.gd: inner_voice / gibber / call_voice

One lineage, three voices. All three are the same synth – a voiced saw source (harmonics summed, each weighted by two vowel _formant() peaks) sequenced into short syllables, with a brief noise "consonant" between. They differ only by register + treatment:

- **inner voice – hers.** The protagonist's inner monologue (S60). Low ~82 Hz, a gentle lowpass, a slow unstable wobble – muffled, behind-the-eyes, deliberately not a babble. This is her.
- **gibber – the night babble.** The grunts' warped psychosis voice (S29), ~150 Hz, full-band, harsher consonants.
- **call – the phone.** The classmate on the wire (S55): the same source an octave up (~300 Hz) through a telephone band-pass, gentle consonants. It's **non-semantic** by design – the game plays a short random-vowel babble per beat (2–4 syllables), not text. The buttons play that; the box

drives only the syllable count from your text's length, in whichever voice you pick.

▷ inner voice - hers

▷ gibber - the night

▷ call - the phone

headphones for the

inner voice (~82 Hz).

i kept the door. it usually is, at this hour.

inner (hers) ▼

▷ say the whole line

length → syllable count; vowels + pitch are

random per syllable, exactly like the game.

[the ported synth](#) ▼ [_formant\(\)](#) ▼ [the three voices' params](#) ▼

An expansion - text-articulated animalese NOT IN THE GAME

An expansion, not the canon. Unlike D1, this maps your text to vowels + formants so it reads as almost words, and it sours on the bleed - for the desk voice-lab, where a random babble isn't enough. One **blip per syllable**. Each blip is a little pitched vowel: a sawtooth at the syllable's note, shaped by two **formant** bandpasses chosen from the syllable's vowel (a e i o u), with a consonant-shaped noise transient on the attack - enough articulation that it reads as almost words without ever being one. Notes walk a **pentatonic**, and a tiny per-blip pitch "bounce" gives the animalese lift. Prosody is punctuation: a statement drifts down, a ? tilts the last blips up, , and . are breaths. And the whole thing rides the bleed - see the law below. Deterministic from the text, so a given line always sounds the same.

type anything - she'll say it (as gibberish):

The reading room is open. It usually is, at this hour.

Noted. You're in the minutes now.

The hour is kept. Nothing else is.

A clerk of what remains.

Some drawers are locked. The inv...

Not on file.

The reading room is open. It usu...

Circulation: one. Readership: as...

▷ speak - animalese

▷ speak - whisper (the alt)

drive the top bleed slider between

takes - hear her sour.

the animalese engine ▾ one blip, in full ▾ the whisper alt ▾

the two candidates, side by side

animalese

recommended

Pitched formant-blips on a pentatonic; bouncy, cute, legible as feeling. It fights the desk's dry administrative register – the tension is the character. This is the game's Yūrei; the desk borrows it.

whisper

Silent Hill / P.T.

Filtered-noise breath-clusters, no tonal center – the office exhaling. Register-true and safe, but it agrees with the dry desk instead of complicating it. Keep it as the reduced/"sfx-lite" fallback.

the souring law – putting any of her voices on the one dial

The D1 game voices are static per call. To make one sour the way the day+night bleed does: drop the pitch, open the wobble + lowpass, and surface an inharmonic 2.76× partial (her bell's ratio) as rot() rises – the animalese already does this; the same moves port straight onto voiceBuffer().

bleed 0 (spotless / day): bright root, major-ish pentatonic, short bouncy blips, tight clean formants – cute, and wrong for how ordinary the day is. **bleed → 1 (the wrong hour):** the root descends; the scale bends to a minor/b2 set; pitch-jitter creeps in (she's unstable, tearful); the formants smear (Q opens); an inharmonic 2.76× partial (her bell's ratio) and a sub-octave grow underneath (the yūrei under the girl); blips lengthen and ring; a 55 Hz dread bed (her purr's note) fades up under the line. **She never goes silent** – the escalation rots timbre, space and pitch, never the gain. That is trap #20, in a voice.

Wiring it into yurei-assistant.js (the notes that make it drop-in).

- **Canon first.** Her real voice is D1 – voiceBuffer() here is a byte-port of Sfx.gd's inner_voice / gibber / call_voice; call it directly. The animalese + whisper are desk-lab expansions on that lineage.
- **Trigger:** call animalese(text) where the panel appends a desk line – addLine("desk", r.response, link, "speak") – and let the utterance length drive the speak clip (hold speak until the last blip, then idle).
- **Default OFF.** The assistant is silent by design – ship this behind a pref (reuse wrong-hour's sfx, or a new voice toggle in the FX terminal), default 0. Opt-in only.
- **CSP-clean.** The admin/site CSP is script-src 'self': no AudioWorklet-via-Blob, no eval. This engine is oscillators + noise buffers + WaveShaper only – same primitives already in wrong-hour.js. No media-src needed (nothing is fetched).
- **Share the room.** Route the voice bus into wrong-hour's room so she sits in the same reverb, and read the same rot()/bleedNow() for the souring – one dial, whole system.
- **Respectful.** Honour the first-gesture gate (whenReady), prefers-reduced-motion (fall

back to whisper or to text-only), and keep it dismissible. Rotate/duck the bed under her so she's never fighting the drone.

C · Couple it – one hand on everything

The point isn't a drawer of effects; it's that **one number conducts sound and vision at once**. "Start the room" plays a coupled hum+bed; then the console slider, in real time, raises and detunes the hum, opens the noise filter, deepens the colour, widens grain and aberration, curdles the bell, and sours her voice – all from the same value. Start the room, move the slider, and let her say a line at each setting.

she says a line, at this bleed

the same words at bleed 0 and bleed 0.85 are two different characters.

Wiring it in. Keep bleed a single source (clock, tension, or a per-day cookie). Every audio patch reads it at fire time; every visual loop reads it per frame; the palette reads it on change; the voice reads it per blip. Nothing else needs to know why the site is souring – one dial, and the whole room answers.

– built for wuld.ink, v2. Take any block. The door's open, mostly because none of us figured out how to close it.